

Comments and Possible Extensions for the Main Author

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1 Overall assessment

After working through the details, I think the central idea is conceptually strong:

coarse solver $\longrightarrow$ prolongation $\longrightarrow$ adiabatic refinement $\longrightarrow$ RODEO filtering.

The most distinctive contribution is not UCCSD or RODEO individually, but the multiresolution viewpoint: eigenstate preparation is treated as a continuation problem across a hierarchy of Hilbert spaces.

The multigrid analogy is particularly effective and could perhaps be emphasized even more in the introduction and conclusion.

2 Pairing model as a favorable benchmark

The pairing Hamiltonian is a natural proof-of-principle model because it is transparent, exactly benchmarkable, and physically meaningful.

At the same time, it is unusually favorable to both the variational and refinement parts of the pipeline.

Because seniority is conserved:

- all singles have zero gradient,
- pair-breaking doubles have zero gradient,
- only pair-preserving doubles survive,
- ADAPT-VQE would strongly compress the full UCCSD ansatz.

In addition, the tested instances exhibit:

$$p_{\text{prolonged}} \simeq 1,$$

a nonclosing refinement gap, and a very high RODEO acceptance probability.

This raises a natural question:

How much of the observed efficiency is generic, and how much is due to seniority conservation?

I think the paper would be strengthened, either now or in follow-up work, by testing the refinement strategy on a less symmetric Hamiltonian.

3 Suggested next model: Fermi–Hubbard

The first candidate that comes to mind is the Fermi–Hubbard model,

$$H = -t \sum_{\langle i,j \rangle, \sigma} \left(c_{i\sigma}^\dagger c_{j\sigma} + c_{j\sigma}^\dagger c_{i\sigma} \right) + U \sum_i n_{i\uparrow} n_{i\downarrow}. \quad (3.1)$$

It provides a useful stress test because:

- there is no seniority conservation analogous to the pairing model,
- single-particle and more general correlation effects are relevant,
- the coarse-state overlap may degrade as U/t increases,
- the refinement interpolation may develop a smaller minimum gap,
- ADAPT compression should be less trivial,
- exact diagonalization remains possible on small clusters.

A 2×2 cluster, a short one-dimensional chain, or possibly a 2×3 cluster would already provide a meaningful first test.

Useful benchmark regimes would include:

1. weak coupling,

$$U/t \ll 1,$$

where Hartree–Fock-like states remain reasonable;

2. intermediate coupling,

$$U/t \sim 4,$$

where correlations become substantial;

3. strong coupling,

$$U/t \gg 1,$$

where the low-energy physics approaches an effective spin model.

The key diagnostic would be whether the pipeline remains effective as the prolonged overlap decreases and the minimum interpolation gap becomes smaller.

4 Other possible test models

Other natural candidates are:

- molecular electronic Hamiltonians,
- realistic nuclear shell-model Hamiltonians beyond pure pairing,
- frustrated spin Hamiltonians,
- extended Hubbard models,
- pairing Hamiltonians with explicit seniority-breaking perturbations.

A particularly controlled extension would be to start from the present pairing model and add a tunable symmetry-breaking term,

$$H(\lambda) = H_{\text{pair}} + \lambda V_{\text{break}}, \quad (4.1)$$

then study the pipeline as λ is increased.

This would isolate the effect of losing seniority conservation while keeping the rest of the benchmark structure close to the current model.

5 Positioning relative to ADAPT-VQE

The current discussion correctly states that ADAPT and the present pipeline are complementary.

This point could be made even sharper:

ADAPT-VQE addresses how to construct a compact coarse ansatz,
whereas

refinement + RODEO addresses how to improve a high-overlap state beyond the variational manifold.

Therefore ADAPT is not primarily a competing alternative to the pipeline. It is a possible front-end coarse solver.

The same is true of:

- Hartree–Fock,
- selected configuration interaction,
- tensor networks,
- classical eigensolvers,
- other variational quantum algorithms.

The framework is solver-agnostic, and this is one of its strengths.

6 Clarify the adiabatic assumption

The minimum instantaneous gap,

$$\Delta_{\min} = \min_{s \in [0,1]} [E_1(s) - E_0(s)],$$

is central to the cost of refinement.

In the present model the spectrum is shown numerically to remain open. It may be useful to state explicitly that this is not guaranteed for a generic Hamiltonian.

A short discussion could mention that refinement may become difficult when:

- the coarse and fine Hamiltonians lie in different phases,
- the interpolation crosses an avoided crossing with exponentially small gap,
- the prolongation maps the coarse state into the wrong symmetry sector,
- the embedded subspace has poor overlap with the relevant fine-resolution eigenstate.

This would clarify the scope of the method without weakening the main result.

7 Clarify the cost comparison

The comparison with direct UCCSD-VQE is informative, but it is important to emphasize that the advantage is not shallow depth.

For the reported operating point,

$$C_{\text{refinement}} = 80 \times 296 \times 2 = 47360$$

CNOTs, and the two RODEO cycles add roughly

$$2.3 \times 10^4$$

CNOTs.

The full pipeline therefore costs approximately

$$7.1 \times 10^4$$

CNOTs per attempt, compared with 1312 CNOTs for a replay of the optimized $k = 4$ UCCSD circuit.

The pipeline is therefore much deeper.

What the extra depth buys is:

- freedom from the fixed UCCSD variational floor,
- optimizer-free exponential suppression in the RODEO stage,
- systematic improvability through longer adiabatic evolution, finer Trotterization, and additional filtering cycles.

The paper already says that there is “no free lunch” in coherent depth. I think it would help to frame the method explicitly as a fault-tolerant or long-coherence strategy rather than as a near-term circuit-depth improvement.

8 RODEO-cycle interpretation

The choice $M = 2$ could be explained as an operating-point optimization rather than as a fundamental limit.

One attempt contains two RODEO cycles. The probabilistic overhead is included through

$$C_{\text{accepted}} = \frac{C_{\text{attempt}}}{P_M}.$$

Because the refined input already has very high overlap,

$$P_M \simeq 0.99,$$

the repetition overhead is negligible.

Larger M improves spectral purity but also increases coherent depth and can reduce acceptance. The relevant comparison is therefore not simply fidelity versus M , but expected cost per accepted state versus M .

A small plot of

$$\frac{C(M)}{P_M}$$

against final infidelity could make this trade-off especially clear.

9 Possible numerical additions

For a less symmetric model, I would suggest recording at each resolution:

$$p_{\text{prolonged}},$$

$$\Delta_{\text{min}},$$

$$P_M,$$

$$E - E_{\text{exact}},$$

$$1 - |\langle \psi_{\text{exact}} | \psi \rangle|^2,$$

and

$$C_{\text{accepted}}.$$

It would also be useful to compare several coarse solvers:

- Hartree–Fock,
- UCCSD,
- ADAPT-VQE,
- possibly a small tensor-network or selected-CI calculation.

This would separate the quality of the coarse solver from the intrinsic performance of the refinement and filtering stages.

10 Main recommendation

My strongest suggestion is therefore:

Test the refinement pipeline on a Hamiltonian that does not conserve seniority, with the Fermi–Hubbard model as the most natural first example.

A small exact-diagonalization benchmark would already answer several important questions:

- Does prolongation still produce a useful overlap?
- Does the interpolation remain gapped?
- How rapidly does the adiabatic cost grow?
- Does RODEO remain inexpensive once the refined overlap is lower?
- Does the advantage over a compact adaptive ansatz persist?

Even if the performance deteriorates, the result would be informative because it would identify which component of the pipeline is most sensitive to the loss of symmetry.

11 Concluding perspective

The central message I would emphasize is:

The paper proposes a multiresolution architecture for eigenstate preparation,

not merely another variational ansatz or another spectral filter.

The coarse-state solver is modular. The prolongation and refinement stages transport information across resolutions. RODEO then supplies optimizer-free spectral purification once the overlap is sufficiently large.

That architecture is the most original aspect of the work and, in my view, deserves to remain the main focus of the presentation.